	Patient 1	Patient 2
Age	70	65
Initial PSA (ng/ml)	4.49	5.57
Initial GS	GG1/3+3=6	GG1/3+3=6
Initial cTNM	cT1N0M0	cT1N0M0
HIFU date	2/3/2021	11/11/2021
PSA prior to HDR (ng/ml)	10.5	4.82
Biopsy path for recurrence	GG2/3+4=7	GG2/3+4=7
HDR 1 dose (cGy)	1350	1250
HDR 1 # needles	16	17
HDR 1 D90	120%	109.80%
HDR 1 V100	97%	97%
HDR 1 Urethral Dmax	117%	117%
HDR 1 Rectal D1cc	68%	66%
HDR 1 Rectal D2cc	59%	59%
HDR 2 dose (cGy)	1350	1250
HDR 2 # needles	16	17
HDR 2 D90	112%	107%
HDR 2 V100	95%	94%
HDR 2 Urethral Dmax	115%	120%
HDR 2 Rectal D1cc	66%	63%
HDR 2 Rectal D2cc	59%	56%

Scientific Session: Miscellaneous Proffered Papers I
Thursday, June 19, 2025
4:00 PM - 5:00 PM

MPP01 Presentation Time: 4:00 PM

Clinical Outcomes Following Reduced Dose Iodine-125 Eye Plaque Brachytherapy for Patients with Uveal Melanoma

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Purpose: Prior to the 1980s, the standard treatment of ocular melanoma was enucleation. The Collaborative Ocular Melanoma Study (COMS) clinical trial demonstrated no difference in survival and high rates of local control for patients managed with iodine-125 plaque brachytherapy (85Gy), which allowed for eye preservation, instead of enucleation. The purpose of our study was to evaluate if a reduced dose prescription of 63Gy covering a 2mm radial margin allows for historical high rates of local control (≥90%) while minimizing ocular toxicity.

Materials and Methods: We performed a retrospective analysis of patients treated for uveal melanoma with iodine-125 IsoAid Eye Physics plaques between 11/1/2016 to 3/31/2024, with at least 6 months of follow up. Seed strength was calculated to deliver a prescription of 63Gy to cover a 2mm circumferential margin around the tumor utilizing a 3D planning software. Clinical outcomes including freedom from local recurrence (FFLR), distant metastasis-free survival, and overall survival (OS) were estimated using the Kaplan-Meier method.

Results: Of the 369 patients with uveal melanoma, 54.2% were male. Tumors were most commonly limited to the choroid (n=348, 94.3%), with the remainder being iris (n=7, 1.9%), iridociliary (n=2, 0.5%), ciliary body (n=3, 0.8%), ciliochoroidal (n=7, 1.9%), and iridociliochoroidal (n=2, 0.5%). Median tumor apical height was 4mm (IQR 2.8-5.9) and median longest basal dimension was 12mm (IQR 9-15mm). Of biopsied tumors, 137 (45.4%) had a gene expression profile (GEP) of class 1A, 67 (22.2%) class 1B, 88 (29.1%) class 2, and 10 (3.3%) failed to amplify; 212 (70.4%) were positive for preferentially expressed antigen in melanoma (PRAME), 79 (26.2%) negative for PRAME, and 10 (3.3%) failed to amplify. Median dose was 63.4 Gy (IQR 62.9-63.9) at prescription height, 76.7 Gy (IQR 64.2-93.5) at tumor apex, 23.6 Gy (IQR 13.6-35.1) at the optic disc, and 29.3 Gy (15.6-59.6) at the fovea. With a median follow up of 2.5 years (IQR 1.33-3.8 years), 2-year FFLR was 96.5% and 5-year FFLR was 93.6%. Distant metastasis free survival was 92.9% at 2 years and 78.5% at 5 years. OS was 95.5% at 2-years and 84.5% at 5-years. Median freedom from any radiation toxicity (retinopathy, papillopathy, or maculopathy) was 2.67 years (IQR 1.5-5.9) and was 83.9% at 1-year, 63.5% at 2-years, and 29.5% at 5-years. Enucleation-free survival was 96.4% at 2-years and 90.2% at 5-years. Median baseline visual acuity was 20/32 (IQR 20/25 - 20/63) and median final visual acuity was 20/40 (IQR 20/25 - 20/80). Baseline visual acuity of 20/200 or worse for 32 (8.7%) patients and final visual acuity of 20/200 or worse for 103 (27.9%) patients. After treatment, 20 (5.4%) patients required enucleation—11 for recurrence, 6 for increased intraocular pressure, and 3 for other reasons.

Conclusions: In a large cohort of ocular melanoma patients, local control of ≥90% was maintained at 5-years after reduced dose iodine-125 eye plaque brachytherapy. Post treatment visual acuity of 20/200 or worse was seen in less than a third of patients—compared to visual acuity of 20/200 or worse in about half of patients on the COMS trial. Overall, 63Gy covering a 2mm margin around the tumor allows for high rates of local control while minimizing ocular toxicity.

MPP02 Presentation Time: 4:09 PM

AI-Assisted Development of a Dynamic Physician Grading System for Cosmetic Outcome Evaluation After Brachytherapy for Non-Melanoma Skin Cancer

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Purpose: Evaluating post-treatment cosmetic outcomes after interventions like skin brachytherapy requires standardized, reproducible physician grading systems. However, developing and implementing such a system can be technically challenging, especially for researchers with limited programming expertise. ChatGPT is an artificial intelligence (AI) based large language model designed to understand and generate natural, human-like language text. The platform was rapidly adopted, reaching 1 million users in 5 days, and has been increasingly used in medicine for clinical, educational, and research purposes. This study describes how AI assistance facilitated the iterative development of a Microsoft Access-based grading system, overcoming complex database, VBA scripting, and automation challenges.

Materials and Methods: Our plan was to have six investigators—one attending and one resident from Dermatology, Head & Neck Surgery, and Radiation Oncology—independently evaluate the cosmetic outcomes of 125 non-melanoma skin cancer lesions using before-and-after brachytherapy treatment photos. The first initial AI consultation with ChatGPT involved choosing a database structure. Due to feasibility and patient privacy concerns, we opted for a Microsoft Access-based database hosted locally within our institution's network. Prompted by ChatGPT, we created a photos table to store images and a grading form to store physician scores and comments. We then consulted AI to help choose a validated scale for assessing cosmetic outcomes after interventions and decided to use the Global Aesthetic Improvement Scale (GAIS: 1-very much improved, 2-much improved, 3-improved, 4-no change, 5-worse). We also incorporated our own novel Overall Cosmesis Score (OCS: 1-excellent, lesion or treatment effects not visible; 3-acceptable outcome; 5-poor, worse than before treatment), designed to give an overall gestalt of patient appearance post treatment. A six-option toggle button control (1-5 scale) was implemented using an Option Group. AI-assisted VBA scripting helped us overcome various problems, such as troubleshooting image display issues and automating structured data collection into a Microsoft excel file. GAIS-OCS relatability was assessed using linear regression, and rating differences were analyzed with Kruskal-Wallis tests. Statistical significance was set at $p < 0.05$, and analyses were performed in R.